Teaching notes

This resource is designed to help pupils learn to answer extended response questions effectively.

Students have the word list in front of them plus various coloured pencils.

Use the PPT to reveal the clues one at a time. After each clue is revealed, students decide which words to 'save or scrap', before showing the next clue. Then show students the exam question.

They are then asked to underline certain words using different colours. At this point go over some of their ideas to iron out any misconceptions such as *respiration makes energy.*

When this is done students use prompts to help them plan an answer to the question. A simple mark scheme is revealed and students write an answer to the question.

As an alternative to using the PPT the clues, instructions and mark scheme could be printed on paper and handed out to the students at the appropriate time.

Clue 1

The question is about the carbon cycle.

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Clue 2

The question concerns the role of living things in the carbon cycle and the substances they produce using carbon.

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The question

Describe how living things are involved in the cycle of carbon into and out of the atmosphere.

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Instructions

- Underline the processes involved in red.
- Underline the products and reactants of these processes in yellow.
- Underline any substances which contain carbon in blue.
- Underline the organisms in green.
- Circle words which suggest the breakdown of organic material.

Planning your answer

Now you are ready to plan your answer the question, try to write simple statements which link each process to the products and reactants.

E.g. '*respiration uses glucose*'

Write short statements linking animals and micro-organisms to a circled word.

E.g. '*animals digest food*'

And short statements which link how plants use carbon to some of the blue words.

E.g. '*plants use carbon to make fats*'

Q: Which process is carried out by **all** the organisms?

Remember it is this process which releases CO_2 back into the atmosphere.

For **Level 3** answers you should include the role of micro-organisms and decomposition.

Are there any other words remaining in your 'save or scrap' list which are relevant? Plan how you will use these.

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Writing your answer

You are now fully prepared. Use your statements, your 'save or scrap' list and the mark scheme to write a full answer to the question.

Mark scheme

Level 1	1 – 2 marks: One process named OR one organism named and the carbon compound it uses or produces.
Level 2	3 – 4 marks: A simple explanation which names a process, the organism involved and the carbon compounds used OR produced by that process.
Level 3	5 – 6 marks: A detailed explanation naming two processes and the carbon compounds used and produced AND including the precise role of micro-organisms in the carbon cycle.

Keywords

green plants	nitrogen	water	organic compounds
sedimentation	animals	volcanic activity	combustion
products	respiration	carbon dioxide	daylight
oxygen	decomposers	fats	micro-organisms
photosynthesis	breakdown	lightening	dead material
breathing	proteins	atmosphere	feed
faeces	digest	acceleration	velocity
magnetic field	fossil fuel	decay	ammonia
respire	cellulose	carbohydrates	release
ohms	energy flow	carbon	glucose

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